

Alexandros Alexiou

SENIOR SOFTWARE ENGINEER · SYSTEMS & DISTRIBUTED SYSTEMS

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Summary

Senior Software Engineer specializing in distributed, event-driven systems and backend services. Built and operated high-throughput APIs, Kafka pipelines, batch processing jobs, and data migration workflows across Kotlin/Java microservices, C++ services, and PostgreSQL, applying idempotent processing, reliable data synchronization, and zero-downtime migration and rollout strategies.

Work Experience

Senior Software Engineer

Ioannina, Greece - Remote

BESTSECRET GMBH

Dec. 2023 - present

- Designed and implemented a scalable event-driven AI recommendation platform on AKS, leveraging Spring Boot microservices, Kafka event streams, ONNX Runtime/CLIP image embeddings, and PostgreSQL pgvector similarity search to asynchronously process incoming product images and generate real-time visually matched fashion set-product recommendations.
- Redesigned the Low Stock Notification (LSN) pipeline to process and deliver ~2M notifications per day, incorporating recently added wishlist products into targeting logic while introducing strict per-customer and global caps and dedicated muting tables, reducing end-to-end processing time from hours to minutes and consistently outperforming the previous system across all attribution windows with sustained uplift in attributed visits and session-value orders.
- Built the Back In Stock (BIS) notification service end-to-end in Kotlin and Spring Boot, processing and dispatching ~1.5M back-in-stock notifications per month: Kafka consumer for stock-state transition detection via idempotent upserts, tier-based scheduler (Diamond-first, cascading to Basic) dispatching through Salesforce Marketing Cloud, and retry-with-muting for delivery resilience.
- Rolled out BIS by running both the legacy and new schedulers in parallel, with the new path executing in a non-dispatching mode and writing only to its own subscription store, backed by dual-writes for subscription mutations and a backfill tool for pre-existing subscriptions; the service auto-evicts stale subscriptions and tolerates the eventual consistency between the dual stores via idempotent upserts and Kafka-backed retry.
- Exposed the BIS subscription API via REST and GraphQL (NotifyMe), enabling web and mobile customers to subscribe to out-of-stock variants and receive timely alerts.
- Ported the Product Detail Page from a monolith to an Astro.js frontend with TypeScript, GraphQL, and Vue.js; delivered full E2E and unit test coverage in collaboration with cross-functional teams.
- Migrated Product Detail Page backend to a microservice using Kotlin, Spring Boot, Kafka, and PostgreSQL, improving scalability and maintainability.
- Designed a GraphQL API for mobile apps, reducing over-fetching and improving data exchange efficiency.
- Built a high-performance Kafka consumer to process and persist records in cloud storage, optimizing data consistency and throughput.
- Achieved high test coverage with JUnit and Testcontainers through robust unit and integration test suites.
- Set up Datadog monitoring and alerting for observability and proactive system reliability.
- Conducted k6 performance tests to identify bottlenecks and prepare for peak traffic; added caching on PDP LVP (reducing DB load and latency for high-traffic product reads) and a read replica for the wishlist service (offloading read-heavy wishlist queries from the primary and enabling horizontal read scaling during flash sales).
- Maintained the monolithic application by resolving bugs and delivering incremental enhancements.

Software Engineer

Ioannina, Greece - Remote

TELEPOST – PNEUMATIC TUBE SYSTEMS

Mar. 2025 - Jul. 2026

- On-call for live customer installations: diagnosed and recovered stuck transmissions, jammed routing, and failures after power outages and voltage spikes, then fed the recurring failure patterns back into product hardening and test coverage.
- Led the redesign of the desktop schematic editor, turning it into a CAD-style tool that lets engineers lay out and modify pneumatic tube system configurations with direct manipulation, full undo/redo, interactive pipe drawing and editing, snap-to-grid behaviour, and clear separation between editable and powered-on modes, dramatically shortening the time to design or reconfigure an installation on site.
- Migrated historical statistics storage from per-date text files to an indexed database, eliminating the multi-minute load times engineers previously hit when inspecting large deployments and enabling fast date-range queries, aggregations, and on-the-fly filtering inside the Statistics view; the rollout was idempotent and reported any unparseable legacy rows for review so no historical data was silently lost.
- Built a PDF report generator for the Statistics view so engineers can export totals, individual events, sent/received summaries, and zone breakdowns directly from the app, making it trivial to share deployment reports with the team and customers without screenshotting or copy-pasting; added proper Greek/Unicode font support so reports render correctly for local installations.
- Designed and delivered the Station Service-Limit-Reached feature: each station is now tracked against a configurable per-zone service threshold, with a clear on-canvas service indicator and a one-click reset from the device properties dialog after maintenance, allowing technicians to schedule preventive servicing before stations actually fail and reducing unplanned downtime on customer sites.
- Hardened reliability by persisting per-station usage counters so service progress survives application restarts and power cycles, and by raising test coverage across the new features and the realtime emulator to catch regressions before they reach customer installations.
- Contributed to platform reach and field-upgrade safety by porting the build to macOS and hardening the Windows installer so customer licence keys and user settings are preserved across product upgrades.

Software Engineer

Ioannina, Greece - Hybrid

TEAMVIEWER GERMANY GMBH

Jan. 2023 – Dec. 2023

- Led the extension of TeamViewer's Web API by integrating new backend functionality into existing C++ microservices, exposing scripting capabilities previously limited to the Management Console UI.
- Enhanced and maintained core C++ backend services and their C# integrations, working with C++/CLI and a custom binary serialization protocol to support reliable cross-language communication.
- Resolved technical support issues across C++ service logic, cross-language integrations, and protocol handling, diagnosing production behavior and delivering targeted fixes.
- Implemented Chat Access Control features on macOS by modifying C++ connection and session handling modules to apply user-defined visibility rules during remote sessions.
- Developed migration features for the transition to TeamViewer Remote, modifying C++ service logic to support seamless client switching, including session persistence and UI fallback through auto-restart logic.

Junior Software Engineer

Ioannina, Greece - Hybrid

TEAMVIEWER GERMANY GMBH

May. 2022 – Jan. 2023

- Integrated an acquired IoT solution into TeamViewer's existing architecture by porting functionality from its Java microservices and Node.js gateway to a C#/.NET service layer, integrating with native C++ components through C++/CLI and the TeamViewer communication stack.
- Built a Java-based MongoDB-to-Microsoft SQL Server migration and synchronization pipeline, introducing a new MongoDB document model with an outbox for live changes and a continuously running service to replay them to SQL, while a separate migration service backfilled historical data; used chunked processing, retries with exponential backoff, idempotent writes, and persistent failure tracking and replay to ensure zero data loss during cutover.
- Maintained and optimized the IoT client/frontend stack by diagnosing communication issues, optimizing binary protocol parsing and data handling, reducing redundant React.js requests, and delivering new UI features.

Software Development Intern

Ioannina, Greece - Hybrid

TEAMVIEWER GERMANY GMBH

Oct. 2021 – Apr. 2022

- Developed features for TeamViewer's WebAssembly-based client using C++, Emscripten, React.js, and TypeScript, optimizing remote session logic for browser execution.
- Implemented C++/JavaScript interoperability using Emscripten compiler tooling, exposing C++ data and functionality to React.js/Redux through JavaScript-to-C++ and C++-to-JavaScript callbacks for bidirectional communication between the WebAssembly client and frontend.
- Extended TeamViewer's WebAssembly-based browser client by porting functionality from the desktop C++ client and redesigning its UI with React.js and Fluent UI; modified the underlying C++ and C# layers to expose and integrate desktop-client features and user information, including profile images, display names, and license details.
- Maintained and extended End-to-End test automation for C++ service workflows using the Ranorex API, while developing Cypress test suites, including porting existing Ranorex coverage and adding new end-to-end scenarios to validate key client and service workflows.

Skills

Technical

Java · Kotlin · C++ · TypeScript/JavaScript · Python · Spring Boot · Node.js · Kafka · PostgreSQL · GraphQL · React · Vue.js · Astro.js · Docker · Git · GitLab CI · Kubernetes · Helm

Languages

English (Proficient) · Greek (Native)